Singapore School Kinderland
Secondary-2
Computer Science (O
Level Standard: Sections
1.1 to 1.1.3)
Term 1

Class Test-1 (Sections 1.1 to 1.1.3: Data Representation & Hexadecimal)

Date: _____ **Name: _____**

Total marks: 15 **Marks obtained: _____**

Teacher's signature: _____ **Parent's signature: _____**

Question 1: Data Representation & Multi-Base Conversions [Total: 4 marks]

O Level Exam Source: Cambridge 2210/12/M/J/24 & 2210/11/M/J/24

(a) State the role of logic circuits and transistors in why modern digital computers represent data using binary values. **[1 mark]**

(b) A 12-bit register contains the binary number:

1 1 0 0 1 0 1 0 0 1 1 1

Convert this 12-bit binary number into:
(i) Hexadecimal notation
(ii) Denary number (Show your calculation) **[3 marks]**

Question 2: Hexadecimal Conversions & Memory Storage [Total: 4 marks]

O Level Exam Source: Cambridge 2210/12/M/J/23 & 0478/12/M/J/23

(a) Convert the denary number 826 into Hexadecimal. Show your successive division working clearly. **[2 marks]**

(b) Convert the hexadecimal value 6E2 into binary format. **[1 mark]**

(c) State the minimum number of bytes required to store the hexadecimal number 1A4F in computer memory. Explain your answer. **[1 mark]**

Question 3: Media Access Control (MAC) & IP Addresses **[Total: 4 marks]**

O Level Exam Source: Cambridge 2210/12/M/J/24 & 2210/12/O/N/23

A network device has the following Media Access Control (MAC) address:

```
4 A : 5 B : 6 C : 7 D : 8 E : 9 F
```

(a) (i) Explain what the first 6 hexadecimal digits (4A:5B:6C) represent. **[1 mark]**

(a) (ii) Explain what the last 6 hexadecimal digits (7D:8E:9F) represent. **[1 mark]**

(b) Distinguish between a Universally Administered MAC Address (UAA) and a Locally Administered MAC Address (LAA). **[1 mark]**

(c) IPv6 addresses are written using hexadecimal notation (e.g. 2001:0db8:85a3::8a2e:0370:7334). Explain why hexadecimal is used for IPv6 addresses instead of 128 binary bits. **[1 mark]**

Question 4: Memory Dumps & Error Diagnostics **[Total: 3 marks]**

O Level Exam Source: Cambridge 2210/11/O/N/23 & 2210/12/M/J/24

During program debugging, an operating system outputs a Memory Dump displaying RAM contents:

```
Address: 0x00401A     Data: 4E  6F  74  20  46  6F  75  6E  64
```

(a) Explain what is meant by a Memory Dump and describe how it helps software developers debug programs. **[2 marks]**

(b) State one other practical application of hexadecimal in computer systems (other than MAC addresses, IP addresses, and memory dumps). **[1 mark]**